

# Model Validation Report — Equity Selection Signals

|                            |                                                                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Models under review</b> | gbm (deployed Monte-Carlo ranking), gbm_expected (its closed-form expectation), momentum (12-1 benchmark rule), clam_orig / clam_2021 (CNN-LSTM-Attention forecaster) |
| <b>Model owner</b>         | Long_Term_Trading / Quant_Model_Research (developer: the author, in an earlier role)                                                                                  |
| <b>Validator</b>           | Signal_Validation — independent re-implementation; no model code modified or tuned                                                                                    |
| <b>Data</b>                | 3,000 US equities (top by market cap, snapshot 2026-09-15), daily prices 2013-01-02 → 2026-09-14, weekly rebalance 2015-01-02 → 2026-09-12 (612 weeks)                |
| <b>Report date</b>         | 2026-09-15 (round 1) · 2026-09-16 (round 2 revalidation, §12)                                                                                                         |
| <b>Standard followed</b>   | SR 11-7 / OCC 2011-12 structure: purpose, data, methodology, results, findings, limitations, monitoring, conditions of use, decision                                  |

## 1. Executive summary and decision

| model                   | Q1 overfit tests passed       | Q2 current monitoring status       | Q3 gate benefit (OOT)                      | decision                                                                                         |
|-------------------------|-------------------------------|------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------|
| momentum                | 2 / 3 (fails Deflated Sharpe) | AUC 0.506, PSI 0.25 (watch)        | max DD −29.7% → −22.0%, CAGR 35.5% → 23.7% | <b>Conditionally approved</b> as a benchmark only, under §9 conditions                           |
| gbm_expected            | 0 / 3                         | AUC 0.500, calibration slope −0.03 | gate off 63% of weeks                      | <b>Not approved</b> — no evidence of skill                                                       |
| gbm (deployed)          | 0 / 3                         | AUC 0.502                          | gate off 92% of weeks                      | <b>Not approved</b> — implementation defect (F1)                                                 |
| clam_orig               | —                             | —                                  | —                                          | <b>Not approved</b> — cannot be validated (F3, F4)                                               |
| gbm_weekly (round 2)    | 2 / 3 (fails Deflated Sharpe) | AUC 0.507, calibration slope 0.74  | gate destroys return (CAGR 19.3% → −2.6%)  | <b>Not approved</b> — real but does not beat the momentum benchmark (§12)                        |
| clam_2021 (round 2)     | 0 / 3                         | AUC 0.502                          | gate off 92% of weeks                      | <b>Not approved</b> — original methodology retrained with an honest cut-off shows no skill (§12) |
| clam_weekly_* (round 2) | 0 / 3                         | AUC 0.50                           | gate off 61–92% of weeks                   | <b>Not approved</b> — two redevelopment variants, no evidence of skill (§12)                     |

Across two rounds and nine candidate variants, none is approved for production use; the untuned momentum rule is retained as the benchmark that any future model must beat. The round-2 redevelopment (§12) confirmed that the GBM defect was an implementation error (fixed model passes the noise tests) and that the CLAM approach does not produce a usable weekly signal even after its data-construction defect is repaired. The most valuable outputs are five findings (§7) that came from reading and reproducing the deployed code rather than from any single statistic.

## 2. Purpose and scope

The model owner uses these signals to rank US equities weekly and hold the top names. This review asks the three questions a second-line function must answer before approving a model:

1. Is the backtested performance real, or an artefact of search and noise? (§5.1)
2. Is the signal still valid today, and how would we know when it stops being valid? (§5.2)
3. What is the rule for switching it off, and does that rule help out-of-time? (§5.3)

Out of scope: portfolio construction beyond equal-weight top-50, risk limits, and any change to the models themselves. The validator's job is to accept, reject, or condition — not to redevelop.

### 3. Models under review

| model        | as deployed                                                                                                                        | validator's reproduction                                                                                                                              |
|--------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| gbm          | Fit $\mu$ , $\sigma$ on 504 daily log returns; simulate <b>one</b> 65-step GBM path; score = $\text{mean}(\text{path})/S_0 - 1$    | Ported verbatim from <code>stats_model_process.get_gbm_path_simulation</code> , fixed seed per date                                                   |
| gbm_expected | —                                                                                                                                  | $E[\text{score}] = \text{mean}_i \exp(\mu \cdot t) - 1$ : same model without the Monte-Carlo draw. Added by the validator to measure the draw's noise |
| momentum     | —                                                                                                                                  | $P[t-21] / P[t-252] - 1$ . Textbook 12-1 momentum; no parameter was chosen by looking at the data                                                     |
| clam_orig    | Keras CNN+LSTM+Attention, 252-day input of 5 log-diff features, 65-day output; score = $\exp(\sum \text{predicted dlog High}) - 1$ | Weights and scaler loaded as shipped; inference code ported from <code>clam_inference.py</code>                                                       |
| clam_2021    | —                                                                                                                                  | Same training script, training window cut at 2021-12-31, to obtain an honest out-of-time period                                                       |

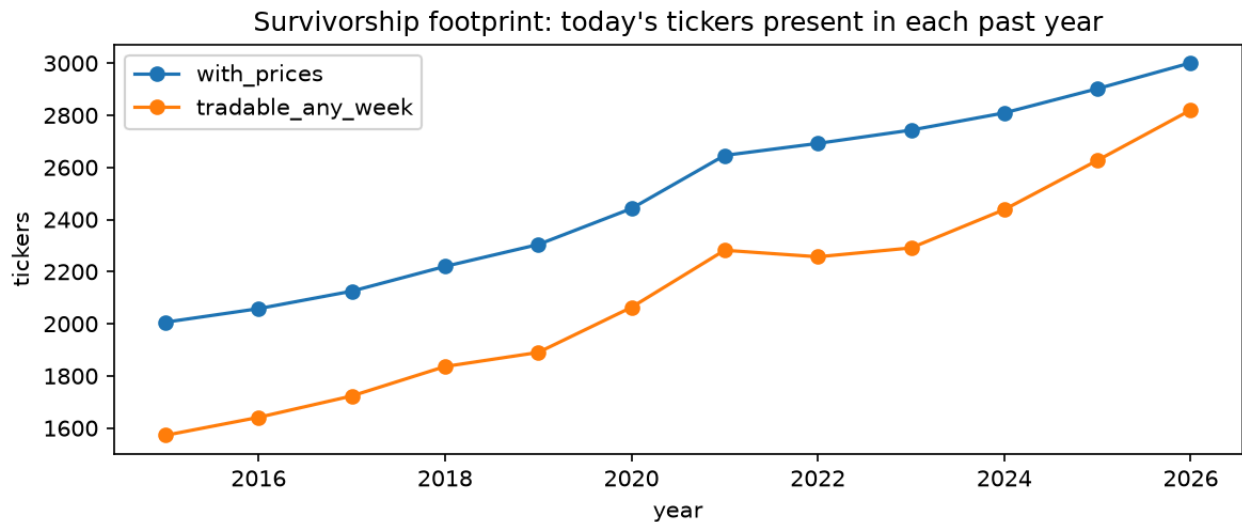
All models are evaluated with one backtest engine: weekly rebalance at Friday close, long-only, top-50 equal weight, 10 bps one-way cost on traded weight, liquidity filter applied before ranking. Performance is judged on **active return versus the equal-weighted tradable universe**, so that survivorship and universe effects shared by all candidates cancel.

### 4. Data

| item            | value                                                                                   | source                                   |
|-----------------|-----------------------------------------------------------------------------------------|------------------------------------------|
| Universe        | 3,000 largest US common equities, one share class per company                           | Yahoo screener, snapshot 2026-09-15      |
| Prices          | 8,104,194 daily OHLCV rows, 3,001 tickers (incl. SPY)                                   | Yahoo Finance                            |
| Rebalance panel | 1,492,706 (week, ticker) rows, 612 weeks                                                | <code>sql/queries/build_panel.sql</code> |
| Tradable filter | close $\geq$ \$5 and 20-day average dollar volume $\geq$ \$5M; 74% of ticker-weeks pass | <code>config.py</code>                   |
| Storage         | DuckDB; every derived table is reproducible from <code>sql/</code> and <code>sv/</code> | <code>sql/schema.sql</code>              |

**Data quality.** Three tickers (CBIO, NFE, DEC) carry zero or negative adjusted closes on 3,909 rows — a Yahoo dividend-adjustment artefact. They are kept in `prices` as received and excluded at the panel and feature layer. One rebalance week (2026-09-14) is a single-day partial week and carries no forward return.

**Survivorship.** Only 2,006 of today's 3,000 tickers have prices in 2015 (1,572 tradable); the count rises monotonically to 3,001 in 2026 (fig. 1). Companies delisted before the snapshot are absent. This inflates absolute returns for every candidate in the same direction; it is the reason all decisions are made on active return against the same universe, and it is listed as a limitation (§8).



## 5. Methodology

### 5.1 Q1 — overfitting tests (`sv/validation/overfit.py`)

| test                                                                          | guards against                                         | statistic                                                   | pass rule     |
|-------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------|---------------|
| Stationary block bootstrap (Politis-Romano, mean block 13 weeks, 5,000 draws) | trusting a point estimate of Sharpe                    | 95% CI of annualised active Sharpe                          | CI excludes 0 |
| Deflated Sharpe Ratio (Bailey & López de Prado 2014), N = 3 trials            | choosing the best of several tries; non-normal returns | P(true SR > expected max SR of 3 noise strategies)          | $\geq 0.95$   |
| Cross-sectional permutation null (500 shuffles of scores within each week)    | the signal–return link being coincidental              | share of permuted Sharpes $\geq$ observed, on gross returns | $p < 0.05$    |

The permutation null is evaluated on **gross** returns. A shuffled signal turns the book over  $\sim 100\%$  a week; net of costs the null would sit at Sharpe  $\approx -1.9$  and any persistent signal would pass on cost savings alone. This was corrected during the review.

### 5.2 Q2 — monitoring (`sv/validation/monitoring.py`, `sql/queries/psi_score.sql`, `auc_by_year.sql`)

For each model and rebalance date, on the trailing 52 weeks ending the week before the date: `psi_score` (score distribution vs the first 104 weeks, 10 development-sample deciles), `psi_input` (trailing-252d realised vol, the regime variable), `auc_1w` / `ks_1w` (discrimination of next-week direction vs the cross-sectional median), `auc_13w` / `ks_13w` / `calib_slope` at the models' 65-day horizon (lagged 13 weeks so no target is used before it is known), `rolling_sharpe` and `active_drawdown` at portfolio level. PSI and AUC are recomputed per year in pure SQL as a cross-check.

### 5.3 Q3 — kill switch (`sv/validation/gate.py`)

A rule-based gate, thresholds fixed before the out-of-time window was examined:

| rule                         | threshold | rationale                                        |
|------------------------------|-----------|--------------------------------------------------|
| <code>psi_score</code>       | $> 0.25$  | industry convention for "population has shifted" |
| <code>rolling_sharpe</code>  | $< 0$     | trailing year of negative active return          |
| <code>auc_1w</code>          | $< 0.50$  | no discriminatory power over the trailing year   |
| <code>active_drawdown</code> | $< -15\%$ | relative drawdown beyond tolerance               |

Any breach switches the model to cash for the coming week; each flip pays the full one-way cost. Portfolio-level metrics are lagged one week so the gate sees only information available on the decision date. Champion (ungated) and challenger (gated) are compared from 2022-01-03 (246 weeks).

## 6. Results

### 6.1 Full-sample backtest, 2015-01 → 2026-09

| strategy              | CAGR  | vol   | Sharpe | max DD | avg weekly turnover |
|-----------------------|-------|-------|--------|--------|---------------------|
| momentum              | 27.0% | 33.8% | 0.88   | −39.9% | 15%                 |
| gbm_expected          | 18.8% | 32.4% | 0.70   | −55.2% | 10%                 |
| gbm (deployed)        | 3.9%  | 27.9% | 0.28   | −46.2% | <b>90%</b>          |
| universe equal-weight | 12.9% | 20.4% | 0.70   | −40.2% | —                   |
| SPY                   | 13.6% | 16.6% | —      | —      | —                   |

### 6.2 Q1 — overfitting tests (active return vs universe)

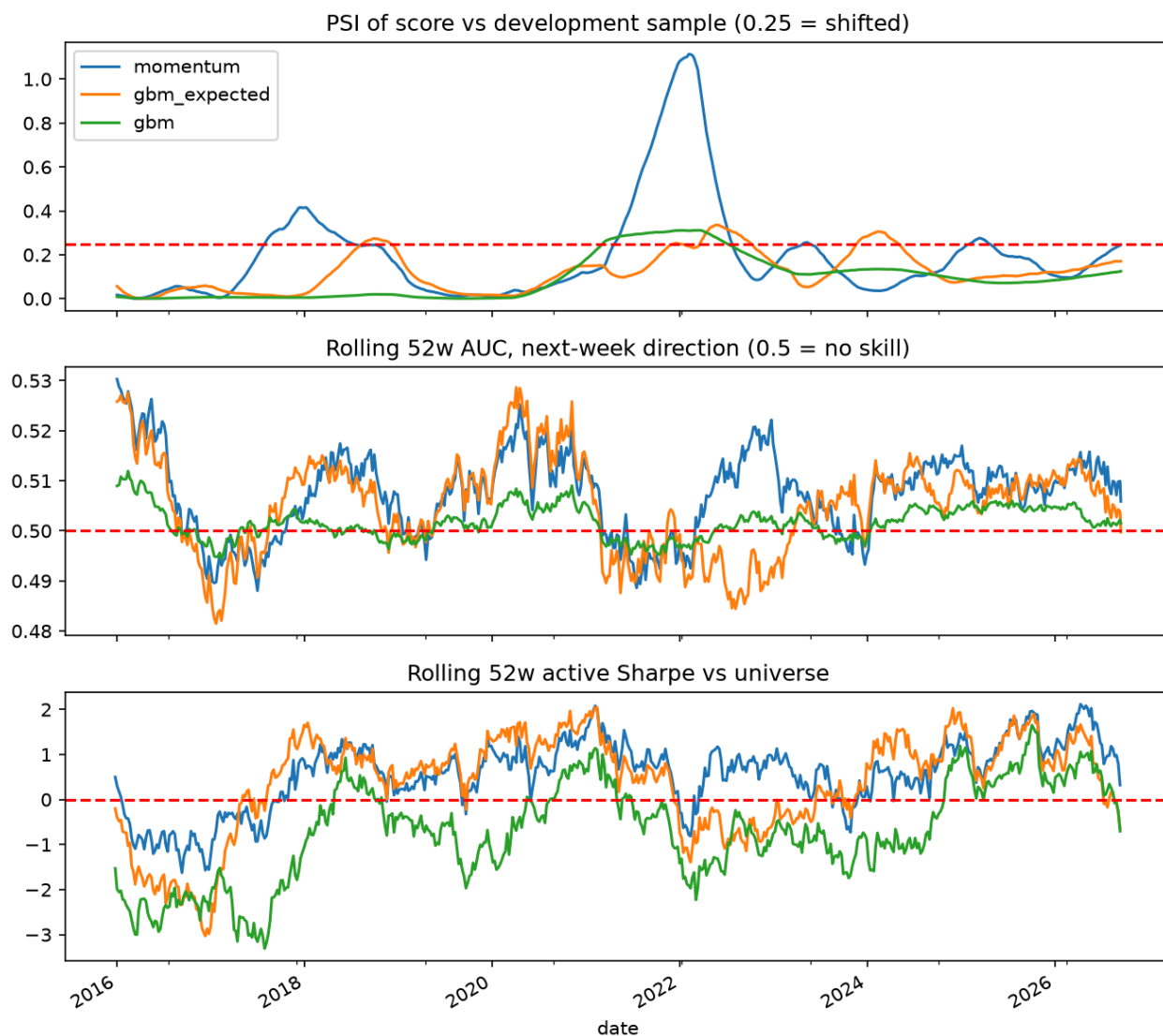
| model        | bootstrap SR [95% CI] | verdict | DSR  | verdict | permutation p | verdict |
|--------------|-----------------------|---------|------|---------|---------------|---------|
| momentum     | 0.65 [0.18, 1.10]     | PASS    | 0.68 | FAIL    | 0.006         | PASS    |
| gbm_expected | 0.37 [−0.17, 0.89]    | FAIL    | 0.31 | FAIL    | 0.086         | FAIL    |
| gbm          | −0.50 [−1.08, 0.05]   | FAIL    | 0.00 | FAIL    | 0.278         | FAIL    |

With only three candidates, the expected maximum Sharpe of pure-noise strategies is already 0.51 annualised; momentum's 0.65 does not clear it with 95% confidence. This is the trading analogue of peeking in sequential testing: every additional candidate raises the bar.

### 6.3 Q2 — monitoring, latest reading (week of 2026-09-12)

| metric                   | momentum        | gbm_expected  | gbm           |
|--------------------------|-----------------|---------------|---------------|
| psi_score                | 0.246 (watch)   | 0.171 (watch) | 0.125 (watch) |
| psi_input (realised vol) | 0.611 (shifted) | 0.611         | 0.611         |
| auc_1w                   | 0.506           | 0.500         | 0.502         |
| auc_13w                  | 0.515           | 0.510         | 0.502         |
| calib_slope (13w)        | —               | −0.03         | 0.01          |

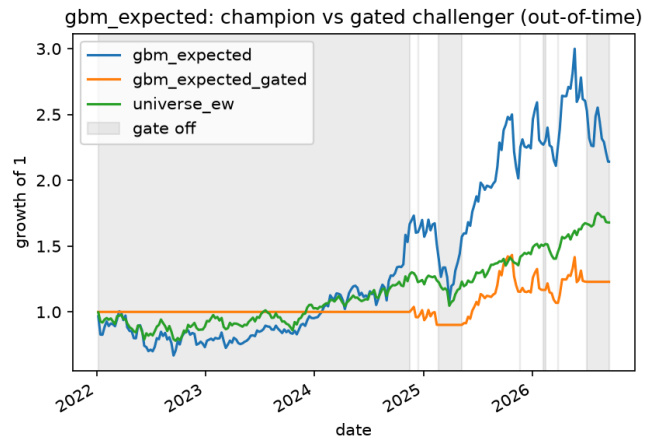
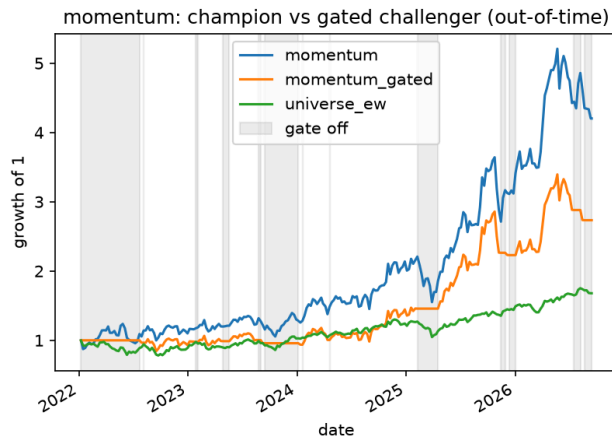
No model shows discriminatory power above 0.53 AUC in any calendar year 2015-2026 (SQL cross-check, `notebooks/02_monitoring.ipynb`). The predicted-return models are uncalibrated: a slope of −0.03 means a higher predicted 65-day return does not translate into a higher realised one. The volatility regime has shifted well beyond the development sample (PSI 0.61), which applies to all candidates equally.



#### 6.4 Q3 — champion vs challenger, out-of-time 2022-01 → 2026-09

| model                 | CAGR champion → challenger | max DD champion → challenger | Sharpe champion → challenger | weeks gated off |
|-----------------------|----------------------------|------------------------------|------------------------------|-----------------|
| momentum              | 35.5% → 23.7%              | -29.7% → -22.0%              | 1.02 → 0.88                  | 48%             |
| gbm_expected          | 17.5% → 4.4%               | -37.5% → -25.4%              | 0.63 → 0.33                  | 63%             |
| gbm                   | 5.9% → 0.0%                | -34.7% → 0.0%                | 0.34 → n/a                   | 92%             |
| universe equal-weight | 11.6%                      | -21.6%                       | 0.67                         | —               |

The gate does what it is specified to do — it cuts maximum drawdown by 26% (momentum) and 32% (gbm\_expected) — but it costs more return than it saves risk: Sharpe falls in both cases. The dominant trigger is active\_drawdown, which fires late. As specified, the kill switch is a drawdown limiter, not a skill detector; a validator would not approve it as the sole control.



## 7. Findings

| #  | model              | severity | finding                                                                                                                                                                                                                                                                                                                                      | evidence                                                  |
|----|--------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| F1 | gbm                | High     | Scoring by a single Monte-Carlo path adds noise of order $\sigma/\sqrt{T}$ that destroys the ranking: weekly turnover 90% vs 10%, CAGR 3.9% vs 18.8% for the identical model's expectation. The defect is in the implementation, not the model.                                                                                              | median weekly Spearman(single path, expectation) = 0.24   |
| F2 | gbm / gbm_expected | Medium   | $E[\text{score}]$ depends on $\mu$ only and exp is monotone, so the "GBM simulation" ranks stocks by 2-year mean log return. The Monte-Carlo machinery adds no information.                                                                                                                                                                  | Spearman(expected score, trailing mean log return) = 1.00 |
| F3 | clam_orig          | Critical | Training sequences are cut from 94 tickers concatenated and <b>sorted by date</b> : a 252-row window covers ~2.7 calendar days of 94 different stocks, while inference feeds 252 days of one stock. The network was never trained on the input it receives in production.                                                                    | code review of clam_model.py vs clam_inference.py         |
| F4 | clam_orig          | Blocking | Weights dated 2025-08-22: every backtest week before that is in-sample. 56 out-of-time weeks remain, i.e. ~4 independent 65-day horizons — too few for any statistical statement.                                                                                                                                                            | panel date count after 2025-08-22                         |
| F5 | momentum           | High     | Entering one session after the rebalance close removes roughly half the out-of-time CAGR (35.5% → 19.2%). The excess return is concentrated in the first session after each rebalance (top-50 first-session return 11–49% annualised in most years vs 4–8% for the universe). The backtest is only valid under close-of-rebalance execution. | ret_fwd_1w_lag1 sensitivity, notebooks/04_findings.ipynb  |
| F6 | all                | Low      | Yahoo returns negative adjusted closes for 3 tickers; without filtering,                                                                                                                                                                                                                                                                     | notebooks/00_data_quality.ipynb                           |

| #  | model     | severity | finding                                                                                                                            | evidence |
|----|-----------|----------|------------------------------------------------------------------------------------------------------------------------------------|----------|
|    |           |          | momentum scores become infinite and monitoring crashes.                                                                            |          |
| F7 | validator | Process  | The original permutation test compared <i>net</i> returns and was biased toward passing any low-turnover signal. Corrected (§5.1). | §5.1     |

## 8. Limitations

1. **Survivorship bias** — unavoidable with free data; quantified in §4 and fig. 1. Absolute returns of every candidate are overstated; relative conclusions are more robust than absolute ones.
2. **Single universe snapshot** — market-cap ranks are as of 2026-09-15; the 2015 universe would have contained different names.
3. **Execution assumptions** — trades at the rebalance close at 10 bps one-way; no market impact, no borrow, no capacity constraints. F5 shows results are sensitive to this.
4. **Multiple-testing count** — the DSR uses  $N = 3$  declared candidates. Any undocumented variants tried during development would raise the bar further.
5. **Deep-learning candidate** — `clam_orig` could not be validated on statistical grounds (F4) and `clam_2021` was not yet scored on the validation machine at the report date. The CLAM decision rests on code review (F3), which is sufficient for non-approval.
6. **Costs** are a flat assumption; small-cap names in the tradable set will trade wider than 10 bps.

## 9. Conditions of use and prohibited uses

Applies to `momentum`, the only conditionally approved model:

- Approved **as a benchmark** for evaluating future models, not as a standalone production signal, until the Deflated Sharpe test is passed on a longer sample or with fewer competing candidates.
- Use only within the tradable universe ( $\text{close} \geq \$5$ ,  $\text{ADV} \geq \$5\text{M}$ ). Scores outside it are undefined for approval purposes.
- Execution must occur at or within the rebalance session close. Any process that enters the following day is **out of scope** (F5).
- Do not rely on the kill switch as the sole risk control; it limits drawdown after the fact and reduced Sharpe out-of-time (§6.4).
- Not approved for horizons other than one week, for leverage, or for short positions.

Prohibited outright: `gbm` in its deployed single-path form (F1); `clam_orig` for any decision (F3, F4).

## 10. Ongoing monitoring plan

| metric                                             | frequency | amber | red  | action on red                                             |
|----------------------------------------------------|-----------|-------|------|-----------------------------------------------------------|
| <code>psi_score</code> vs development sample       | weekly    | 0.10  | 0.25 | gate off; investigate feature drift                       |
| <code>auc_1w</code> (trailing 52w)                 | weekly    | 0.51  | 0.50 | gate off; escalate after 4 consecutive weeks              |
| <code>rolling_sharpe</code> (trailing 52w, active) | weekly    | 0.25  | 0.00 | gate off                                                  |
| <code>active_drawdown</code>                       | weekly    | −10%  | −15% | gate off; review sizing                                   |
| <code>psi_input</code> (volatility regime)         | weekly    | 0.25  | 0.50 | flag: all candidates are outside their development regime |

| metric                        | frequency                          | amber | red | action on red        |
|-------------------------------|------------------------------------|-------|-----|----------------------|
| full re-validation (Q1 tests) | annually, or after any code change | —     | —   | re-issue this report |

All metrics are written to `monitoring` and `gate_decisions` in the DuckDB file by the same code that produced this report, so the monitoring pack is a query, not a re-implementation.

## 11. Recommendations to the model owner

1. **gbm**: replace the single Monte-Carlo path with the closed-form expectation (or many paths); then recognise via F2 that the model is a trailing-mean-return rank and decide whether that is the intended signal.
2. **clam**: rebuild training sequences per ticker so training and inference share an input structure; retrain with a fixed cut-off date and a held-out period of at least three years before resubmitting.
3. **momentum**: document the execution assumption; investigate the first-session concentration (F5) before any live use.
4. **Kill switch**: add a leading indicator (e.g. AUC deterioration with a shorter window) alongside the drawdown rule, and re-test champion vs challenger.

## 12. Round 2 — revalidation after redevelopment (2026-09-16)

Following §11, the model owner redeveloped both rejected models for a **weekly** horizon. The validator's role in round 2 was unchanged: same backtest engine, same three questions, same out-of-time window, and every variant the developer tried is counted as a trial in the Deflated Sharpe Ratio ( $N\_TRIALS\_DSR = 11$ : 3 round-1 candidates + 6 GBM variants + 1 CLAM variant retained after 3 attempts + the `clam_2021` twin).

The `clam_2021` twin — the **original** CLAM training code and 94-ticker universe, unmodified, with the training window cut at 2021-12-31 — was also trained and scored in this round (it could not be run on the developer's laptop in round 1). It answers a question the redeveloped models cannot: does the deployed methodology itself have skill once it is given an honest out-of-time period?

### 12.1 What the developer changed

| model                                  | change                                                                                                                                                                                    | developer-side selection (development sample < 2022 only)                                                            |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <code>gbm_weekly</code>                | horizon 65 → 5 days; the single Monte-Carlo path replaced by the closed-form expectation (F1); lookback re-chosen                                                                         | 6 variants (lookback 63 / 126 / 252 × expected / prob-up); best development Sharpe → <code>gbm_w_126_expected</code> |
| <code>clam_weekly_cs_demeaned</code>   | sequences built per ticker (F3); training cut at 2021-12-31 (F4); horizon 5 days; scalar target = next-5-day log return minus that week's cross-sectional mean; top-500 training universe | validation rank IC 0.014                                                                                             |
| <code>clam_weekly_cs_rank_small</code> | as above with a within-week percentile-rank target and a smaller network                                                                                                                  | validation rank IC 0.016                                                                                             |
| (discarded)                            | raw-return target (IC -0.008); weekly-bar input (collapsed to a constant)                                                                                                                 | —                                                                                                                    |
| <code>clam_2021</code>                 | original <code>clam_model.py</code> unchanged, <code>training_end_date = 2021-12-31</code>                                                                                                | training directional accuracy 0.76 / validation 0.64 (see F11)                                                       |

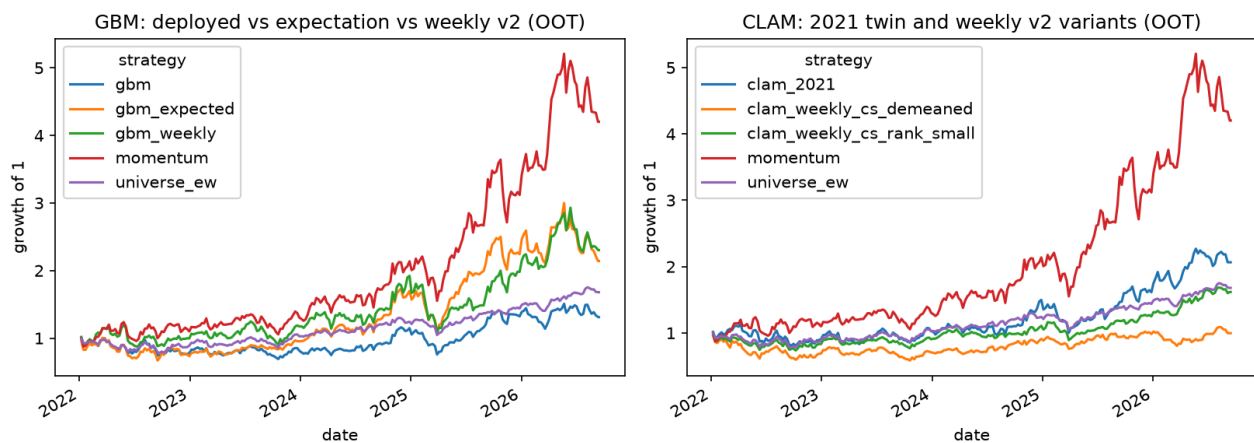
### 12.2 Results

Q1, full sample 2015-01 → 2026-09, active return vs universe (DSR hurdle at  $N = 11$  is  $SR_0 = 0.81$ ):

| model                               | active Sharpe [95% CI] | bootstrap   | permutation p     | DSR       | passed       |
|-------------------------------------|------------------------|-------------|-------------------|-----------|--------------|
| momentum (benchmark)                | 0.65 [0.18, 1.10]      | PASS        | 0.004 PASS        | 0.29 FAIL | 2 / 3        |
| <b>gbm_weekly</b>                   | 0.49 [CI > 0]          | <b>PASS</b> | <b>0.026 PASS</b> | 0.14 FAIL | <b>2 / 3</b> |
| gbm_expected (round 1)              | 0.37                   | FAIL        | 0.080 FAIL        | 0.07 FAIL | 0 / 3        |
| gbm (deployed)                      | −0.50                  | FAIL        | 0.226 FAIL        | 0.00 FAIL | 0 / 3        |
| clam_2021 (original code, 2021 cut) | −0.05 [−0.57, 0.44]    | FAIL        | 0.298 FAIL        | 0.00 FAIL | 0 / 3        |
| clam_weekly_cs_rank_small           | 0.04                   | FAIL        | 0.106 FAIL        | 0.00 FAIL | 0 / 3        |
| clam_weekly_cs_demeaned             | −0.68                  | FAIL        | 0.704 FAIL        | 0.00 FAIL | 0 / 3        |

Out-of-time 2022-01 → 2026-09:

| strategy                  | CAGR  | vol   | Sharpe | max DD | one-day execution lag |
|---------------------------|-------|-------|--------|--------|-----------------------|
| momentum                  | 35.5% | 36.3% | 1.02   | −29.7% | 19.2%                 |
| <b>gbm_weekly</b>         | 19.3% | 36.2% | 0.67   | −43.4% | 8.3%                  |
| gbm_expected              | 17.5% | 35.6% | 0.63   | −37.5% | 11.4%                 |
| clam_2021                 | 16.6% | 30.9% | 0.65   | −33.1% | 9.7%                  |
| clam_weekly_cs_rank_small | 10.7% | 22.5% | 0.57   | −24.1% | 9.1%                  |
| clam_weekly_cs_demeaned   | −0.1% | 26.8% | 0.13   | −39.0% | —                     |
| universe equal-weight     | 11.6% | 19.2% | 0.67   | −21.6% | —                     |



Monitoring: no candidate exceeds AUC 0.53 in any calendar year (SQL cross-check); **gbm\_weekly** is the only model with a meaningful calibration slope at the 13-week horizon (0.74). The gate switches **gbm\_weekly** off 59% of weeks and turns its out-of-time CAGR from 19.3% to −2.6%, confirming the round-1 conclusion that the rule set as specified is not a usable control.

### 12.3 Findings added in round 2

| #  | model             | severity | finding                                                                                                                                                                                                                                                                                                                            |
|----|-------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F8 | <b>gbm_weekly</b> | Medium   | The fixed model is statistically distinguishable from noise (bootstrap and permutation pass) but its out-of-time Sharpe (0.67) equals the equal-weighted universe and trails the untuned benchmark (1.02) with a deeper drawdown. Reason for rejection moves from <i>implementation defect</i> to <i>no value over benchmark</i> . |
| F9 | clam_weekly_*     | High     | After repairing F3/F4 and trying four target/input variants on a 478-ticker, 142k-window training set, the best validation rank IC is 0.016 and out-of-time                                                                                                                                                                        |

| #   | model                 | severity | finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|-----------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                       |          | performance is indistinguishable from the universe. The limitation is the approach — OHLCV sequences alone carry no exploitable weekly cross-sectional signal at this scale — not the implementation.                                                                                                                                                                                                                                                                                                                                         |
| F10 | process               | Low      | Multiple-testing accounting: raising the trial count from 3 to 11 lifts the DSR hurdle from $SR_0 = 0.51$ to 0.81 and fails every candidate including the benchmark. Development iterations must be logged so the validator can count them.                                                                                                                                                                                                                                                                                                   |
| F11 | clam_2021 / clam_orig | High     | The original training code reports directional accuracy of 0.76 (train) / 0.64 (validation), yet the same model scores AUC 0.49–0.50 in every calendar year and an active Sharpe of $-0.05$ . The in-training metric is an artefact of F3: in a window that mixes 94 tickers on the same dates, the "next row" is largely another stock on the same day, so the network learns same-day market direction, which is useless for the single-stock sequences it receives in production. This is how the defect stayed hidden from the developer. |

## 12.4 Decision

| model                                 | round 1                             | round 2                                                                                                                                                                                                                            |
|---------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| momentum                              | conditionally approved as benchmark | unchanged                                                                                                                                                                                                                          |
| gbm → gbm_weekly                      | not approved (F1)                   | <b>not approved</b> — passes noise tests, fails to beat benchmark (F8); may be resubmitted with a documented economic rationale and a smaller trial count                                                                          |
| clam_orig → clam_2021 / clam_weekly_* | not approved (F3, F4)               | <b>not approved</b> — the original methodology has no skill once given an honest out-of-time period (F11), and the repaired version has none either (F9); recommend discontinuing the sequence-model approach for weekly selection |

Recommendation to the owner: stop iterating on CLAM; if GBM-weekly is pursued, its only demonstrated content is trailing mean return (F2), so it should be compared against — or merged into — the momentum benchmark rather than maintained as a separate model.

## Appendix A — assumptions (config.py)

| assumption                             | value                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------------|
| universe size / snapshot               | 3,000 / 2026-09-15                                                                         |
| price history                          | 2013-01-01 → 2026-09-14                                                                    |
| backtest start / rebalance             | 2015-01-02 / weekly, last trading day (Friday)                                             |
| holdings / weighting                   | top 50 / equal                                                                             |
| cost                                   | 10 bps one-way on traded weight                                                            |
| liquidity filter                       | close $\geq$ \$5, 20-day ADV $\geq$ \$5M                                                   |
| momentum                               | 252-day lookback, 21-day skip                                                              |
| GBM                                    | 504-day lookback, 65-day horizon                                                           |
| CLAM                                   | 252-day input, 65-day horizon; original weights through 2025-08-22; retrain cut 2021-12-31 |
| bootstrap                              | stationary, mean block 13 weeks, 5,000 draws                                               |
| DSR trials                             | 3 (round 1) / 11 (round 2)                                                                 |
| monitoring window / development sample | 52 weeks / first 104 weeks                                                                 |

| assumption         | value           |
|--------------------|-----------------|
| out-of-time window | from 2022-01-03 |

## Appendix B — reproduction

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README.md lists the eight commands that rebuild every table and number in this report from a fresh checkout.

Workpapers: notebooks/00\_data\_quality ... 05\_round2; developer side:

Quant\_Model\_Research/clam\_weekly.ipynb, Long\_Term\_Trading/gbm\_weekly.py.